

NAME: _____ JASON _____

ASSIGNMENT 1

Apply the concepts of unit consistency and conversion factors in solving the given problems below.

1. **Convert the following:**

a) 600 mg to kg

$$1 \text{ kg} = 1,000,000 \text{ mg}$$

$$600 \text{ mg} \div 1,000,000 = \mathbf{0.0006 \text{ kg}}$$

b) 10,800 s to h

$$1 \text{ hour} = 3600 \text{ seconds}$$

$$10,800 \text{ s} \div 3600 = \mathbf{3 \text{ hours}}$$

c) 35 mi/h to m/s

$$1 \text{ mile} = 1609.34 \text{ meters}$$

$$1 \text{ hour} = 3600 \text{ seconds}$$

$$(35 \times 1609.34) \div 3600 \approx \mathbf{15.65 \text{ m/s}}$$

d) 80 m^2 to in^2

1 meter = 39.3701 inches

$1 \text{ m}^2 = 1550.0031 \text{ in}^2$

$80 \times 1550.0031 \approx \mathbf{124,000.25 \text{ in}^2}$

e) 20 light year to km

1 light year $\approx 9.461 \times 10^{12} \text{ km}$

$20 \times 9.461 \times 10^{12} = \mathbf{1.8922 \times 10^{14} \text{ km}}$

2. The volume of liquid in a soft drink can is given as 0.355 liters (L) using only the conversions of $1 \text{ L} = 1000 \text{ cm}^3$ and $1 \text{ in} = 2.54 \text{ cm}$, expressing this volume in cubic inches.

1 liter = 1000 cm^3

0.355 liters = 355 cm^3

1 inch = 2.54 cm $\rightarrow 1 \text{ in}^3 = 2.54^3 = 16.387 \text{ cm}^3$

$355 \text{ cm}^3 \div 16.387 \text{ cm}^3/\text{in}^3 \approx \mathbf{21.66 \text{ in}^3}$

3. The gasoline consumption of a small car is advertised as 12 km/L. How many miles per gallon is this?

Standard units: 1 mile = 1.6 km

1 gallon = 3.8 liters

1 mile = 1.6 km

1 gallon = 3.8 liters

$12 \text{ km/L} \times (1 \text{ mile} / 1.6 \text{ km}) \times (3.8 \text{ L} / 1 \text{ gallon}) =$

$12 \times (1 / 1.6) \times 3.8 \approx \mathbf{28.5 \text{ miles/gallon}}$

4. A race car is moving at a speed of 50 mi/h. Express this speed in m/s and km/h.

To m/s: 1 mile = 1609.34 meters

1 hour = 3600 seconds

$50 \times (1609.34 / 3600) \approx \mathbf{22.35 \text{ m/s}}$

To km/h:

1 mile = 1.6 km

$50 \times 1.6 = \mathbf{80 \text{ km/h}}$

In a Nutshell 1

Let's try this critical thinking question based on what you've learned in this lesson.

Question 1. Why is measurement important in our life?

Answer: Measurement is important in our life because it helps us to identify the measure of any objects and helps also with what we are doing in our daily life.

Question 2: What is the importance of using the international systems of units?

Answer: The importance of using the international system of units is to make our measurements clear and accurate, so that we can work easily in our everyday life.